

DATA STRUCTURES WITH ALGORITHMS				
Course Code		25MCA101	Course Type	PCC
Teaching Hours/Week (L: T: P: S)		3:0:0:0	Credits	03
Total Teaching Hours		40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications				
Course Objectives:				
1.	Explain the fundamental concepts of data structures, including primitive and non-primitive data structures, abstract data types, and the basics of algorithm analysis with asymptotic notations			
2.	Implement and manipulate arrays, linked lists, stacks, and queues using C programming to perform operations such as insertion, deletion, traversal, and searching efficiently.			
3.	Apply recursion and stack-based techniques for problem-solving, including evaluating postfix expressions, infix-to-postfix conversion, and solving problems like factorial, Fibonacci, and Tower of Hanoi.			
4.	Analyze and implement binary trees and graphs, perform various tree traversal techniques, and apply graph algorithms such as BFS, DFS, Prim's, Kruskal's, and Dijkstra's algorithms for real-world applications.			
5.	Develop and compare sorting algorithms like Bubble Sort, Quick Sort, Merge Sort, and Shell Sort, and implement searching algorithms including Sequential Search, Binary Search, and Interpolation Search to optimize data retrieval.			
UNIT-I				
Introduction				04 Hours
Introduction to data structure, Primitive and Non-Primitive Data Structures, Abstract data type, Arrays, Pointers				
Algorithm Analysis				04 Hours
Fundamentals of Algorithm Design and Analysis, Analysis Framework, Time Complexity and Space Complexity, Asymptotic Notations: Big O, Omega (Ω), Theta (Θ). Basic efficiency classes,				
UNIT-II				
Lists				08 Hours
Introduction to Linked lists, Types of linked lists, C implementation of Singly Linked Lists, Doubly linked lists, Circular lists.				
UNIT-III				
Stack				06 Hours
Definition and Primitive operations, Representing and implementing stack using Arrays and Linked Lists, Applications of stacks: Evaluating a postfix expression, converting an expression from infix to postfix, Recursion: definition and processes, Implementation of recursion program in C, Examples (Factorial function, Multiplication of natural numbers, Fibonacci sequence, Binary search, Towers of Hanoi problem).				
Queues				02 Hours
Definition and Primitive operations, Implementation of ordinary queues and circular queues using Arrays and Linked lists				
UNIT-IV				
Binary trees				05 Hours

Basic Terminology, Operations on binary trees, Applications of binary trees. Binary tree representation: Node representation of binary tree, Array representation of binary trees, Binary tree traversals and related properties, Threaded binary trees.

Graphs	03 Hours
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Definitions, Representation: Adjacency Matrix, Adjacency List, Graph Traversal Techniques: BFS, DFS. Application of graphs - Greedy Technique, Prim's Algorithm, Kruskal's Algorithm, Dijkstra's Algorithm.

UNIT-V

Sorting	05 Hours
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Exchange Sort: Bubble Sort, Selection Sort, Quick Sort. Divide-and-Conquer: Merge sort. Decrease-and-Conquer - Insertion Sort, Shell sort. Time Complexity Comparison.

Searching	03 Hours
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Sequential search, Indexed sequential search, Binary search, Interpolation Search.

Course Outcomes: At the end of the course student will be able to

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|----|---|
| 1. | Explain the fundamental concepts of Data Structures and the basics of Algorithm Analysis. |
| 2. | Implement Linear Data Structures such as Arrays, Linked lists. |
| 3. | Apply and Implement Recursive and Non-Recursive Techniques. |
| 4. | Analyze and Implement Binary Trees and Graphs |
| 5. | Develop, Implement and Compare different Searching and Sorting Techniques. |

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA101.1	3	3	-	-	-	-	-	-	3	2
25MCA101.2	3	3	3	2	3	-	-	-	3	2
25MCA101.3	3	3	3	3	3	-	-	-	3	2
25MCA101.4	3	3	3	3	3	-	-	-	3	2
25MCA101.5	3	3	3	3	3	-	-	-	3	2

1: Low 2: Medium 3: High

TEXTBOOKS:

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|----|---|
| 1. | Yedidyah Langsam and Moshe J Augenstein and Aaron M Tenenbaum: Data Structures using C and C++, 2 nd Edition, Pearson Education Asia |
| 2. | Reema Thareja – "Data Structures Using C", 2 nd Edition, Oxford University press |
| 3. | Jean-Paul Tremblay: An Introduction to Data Structures with applications, Tata McGraw Hill |
| 4. | Ellis Horowitz and Sartaj Sahni, Fundamentals of Data Structures in C, 2 nd Edition, Universities Press. |
| 5. | Introduction to the Design and Analysis of Algorithms. Anany Levitin, Pearson Education, 2 nd Edition. |
| 6. | Horowitz E., Sahani S., Rajashekharan S: Computer Algorithm, Galgotia Publications. |

REFERENCE BOOKS:

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| 1. | Robert L Kruse : Data Structures and Program Design , 3 rd Edition, Prentice – Hall of India. |
| 2. | Mark Allen Weiss : Data Structures and Algorithm Analysis in C, 2 nd Edition, Pearson Education Asia. |
| 3. | Yashavant Kanetkar – "Data Structures Through C", BPB Publications. |

ADVANCED DATABASE SYSTEMS				
Course Code		25MCA102	Course Type	PCC
Teaching Hours/Week (L: T: P:S)		3:0:0:0	Credits	03
Total Teaching Hours		40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications				
Course Objectives:				
1.	Understand the importance of database management systems and have thorough understanding of terminologies used.			
2.	Implement concepts of relational model using SQL.			
3.	Use the features of PL/SQL to write procedural programs.			
4.	Design the databases and to use different levels of normalization.			
5.	Understand the working of NoSQL, and implement features of MongoDB.			
UNIT-I				
Essentials of Databases and ER Modeling				08 Hours
Introduction, Characteristics of Database approach, Advantage of using DBMS approach, Disadvantages of DBMS. Data Models, Three Schema Architecture, Entity Types, Entity Sets, Attributes and Keys, Relationship types, Relationship Sets, Roles and Structural Constraints, Weak Entity Types, Refining the ER Design.				
UNIT-II				
SQL - The Relational Database Standard :				08 Hours
SQL Data Definition and Data Types, Specifying basic constraints in SQL, DDL, DML, DQL statements in SQL, Aggregate functions, Sub-queries and Co-related sub-queries, LIKE, Inner and Outer Joins, Views in SQL.				
UNIT-III				
Procedural Programming with MySQL:				08 Hours
Introduction stored programs, Writing procedural code, COMMIT, ROLLBACK and save points, Implicit and explicit locking, Stored Procedures, Stored functions, Triggers, Cursors.				
UNIT-IV				
Structured Database Design and Transaction Processing				08 Hours
Informal Design Guidelines for Relation Schema's, Functional Dependencies, Normal Forms (1NF, 2NF and 3NF), Closure sets, Decomposition. Introduction to Transaction Processing, Transaction States, Desirable Properties of Transactions.				
UNIT-V				
Foundations of NoSQL and MongoDB				08 Hours
Introduction to NoSQL, Types of NoSQL databases, Advantages of NoSQL, Comparison of SQL and NoSQL. What is MongoDB? Why MongoDB? Using JSON, Creating or generating unique key, Support for dynamic queries, Terms used in RDMS and MongoDB, Data types in MongoDB, MongoDB Query Language: Inserting, updating and deleting documents, Retrieving documents, Dealing with Null values, Count, Limit, Sort and Skip methods.				
Course Outcomes: At the end of the course student will be able to				
1.	Analyze the characteristics of database systems and apply Entity-Relationship (ER) modeling to design structured databases.			

2.	Construct and manipulate relational databases using SQL statements, including constraints, joins, views, and aggregate functions.
3.	Develop stored procedures, functions, triggers, and cursors in MySQL to implement procedural database programming.
4.	Evaluate functional dependencies and apply normalization techniques to design efficient and consistent relational schemas.
5.	Differentiate between SQL and NoSQL databases, and demonstrate the ability to perform CRUD operations in MongoDB using JSON-based queries.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA102.1	3	-	3	2	3	2	-	3	3	-
25MCA102.2	3	-	3	3	2	3	-	3	3	-
25MCA102.3	3	-	2	3	2	3	-	3	3	-
25MCA102.4	3	-	2	3	2	3	-	3	3	-
25MCA102.5	3	-	3	3	3	3	-	3	3	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Elmasri and Navathe: Fundamentals of Database Systems, 7th Edition 2016
2. Joel Murach: Murach's MySQL, 4th Edition 2023
3. Seema Acharya, Subhashini Chellappan, "Big Data Analytics", 2nd Edition, Wiley, 2016

REFERENCE BOOKS

1. Silberschatz, Korth and Sudarshan: Database Systems Concepts, 6th Edition, McGraw-Hill
2. Raghu Ramakrishnan and Johannes Gehrke : Database Management Systems, 6th Edition, McGraw-Hill
3. Ivan Bayross : Commercial Application Development using Oracle Developer 2000

E Books / MOOCs/ NPTEL

1. NOC : Fundamentals of Database Systems(Course sponsored by Aricent), IIT Kanpur, Dr. Arnab Bhattacharya : <https://nptel.ac.in/courses/106104135>
2. MySQL® Notes for Professionals book: <https://www.goalkicker.com/MySQLBook/>
3. Learning MongoDB: https://www.dbooks.org/learning-mongodb-5605980453/#google_vignette
4. Oracle PL/SQL Language Pocket Reference, 5th Edition: A Guide to Oracle's PL/SQL Language Fundamentals.
<https://www.pdfdrive.com/oracle-plsql-language-pocket-reference-5th-edition-a-guide-to-oracles-plsql-language-fundamentals-d167357800.html>

DATA COMMUNICATION AND COMPUTER NETWORKS			
Course Code:	25MCA103	Course Type:	PCC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits:	03
Total Teaching Hours:	40	CIE + SEE Marks:	50+50
Teaching Department: Master of Computer Applications			
Course Objectives: This Course will enable students to			
1.	To introduce fundamental concepts of data communication, networking architectures, protocols, encoding techniques, and error detection mechanisms for reliable enterprise communication.		
2.	To provide an understanding of data link control mechanisms, flow control, error handling techniques, and key network services like email, DNS, and HTTP for efficient and reliable communication.		
3.	To explore transport layer services and protocols, focusing on process-to-process communication, flow and error control, congestion control, and key transport protocols such as UDP, TCP, and SCTP.		
4.	To understand congestion control techniques, traffic management strategies, and congestion handling mechanisms in packet-switched networks, including TCP and Datagram Congestion Control Protocol (DCCP).		
5.	To introduce security protocols, cryptographic algorithms, and multimedia information processing techniques, including data compression and image/video coding for secure and efficient communication.		
UNIT-I			
Introduction:			02 Hours
Data Communication and Networking for today's Enterprise, A communication Model, Data Communication, Network Types: LAN, WAN, Internet, Internet Architecture.			
Networks Protocols and Architecture:			02 Hours
The Need for a protocol Architecture, A simple protocol Architecture, TCP/IP Protocol Architecture : The TCP/IP layers.			
Data Encoding and Communication Interfaces:			04 Hours
Digital Data & Signals: NRZ, Manchester, Multilevel binary, Biphase, Modulation rate, Differential Encoding, Scrambling Techniques, Analog & Digital Signal Encoding: Modulation Techniques, Pulse Code Modulation, Delta Modulation, Types of errors, Error Detection & Correction: Parity Check, CRC, Forward error correction.			
UNIT-II			
Data Link Control:			04 Hours
Flow Control: Stop and Wait Sliding Window Flow Control; Error Control: Stop and Wait ARQ, GO-Back-N ARQ, Selective-Reject ARQ; High-level Data Link Control: Basic Characteristics, Frame Structure.			

Application Layer:										04 Hours	
SMTP, MIME- Internet Mail Architecture, POP, IMAP, Internet Directory Services- DNS, The DNS Database, DNS operations, Web Access and HTTP: HTTP Overview, Messages, Request Message, Response Messages, Entities.											
UNIT-III											
The Transport Layer:										08 Hours	
Transport Layer Services: Process-to-Process Communication, Addressing: Port Numbers, Encapsulation and Decapsulation, Multiplexing and Demultiplexing, Flow control, Error control, Combination of flow control and error control, Congestion control, Connectionless and connection-Oriented protocol.											
Transport Layer Protocols: Services, Port Numbers, UDP services, TCP Services, TCP Features, Segment, A TCP Connection, State Transition Diagram, SCTP, SCTP services, SCTP Features, Packet format, An SCTP Association, Flow Control and error control.											
UNIT-IV											
Network layer:										08 Hours	
Network layer design issues: Store and Forward packet Switching, Services Provided to the Transport Layer, Implementation of Connection less Service, Implementation of Connection-Oriented Service, Comparison of Virtual Circuit and Datagram Subnets; Routing algorithms: The Optimality Principle, Shortest Path Routing, Flooding, Distance Vector Routing, Link state Routing, Hierarchical Routing, Broadcast Routing, Multicast Routing.											
UNIT-V											
Congestion Control:										08 Hours	
Effects of congestion, Congestion Control, Traffic Management, Congestion Control in Packet Switching Networks, TCP Congestion control, Datagram Congestion Control Protocol											
Course Outcomes: At the end of the course student will be able to											
1.	Demonstrate an understanding of network models, protocols, data encoding, and error correction techniques to analyze and optimize communication systems.										
2.	Analyze and apply data link control protocols, error control mechanisms, and network services like DNS, email, and HTTP to enhance data communication efficiency..										
3.	Analyze and apply transport layer mechanisms, including flow control, error handling, congestion management, and transport protocols (UDP, TCP, SCTP) for efficient data communication.										
4.	Analyze network layer design issues, including packet switching and service implementations, and apply various routing algorithms to evaluate their efficiency and scalability in network communication.										
5.	Analyze and apply congestion control strategies in network communication, including traffic management, TCP congestion control, and DCCP, to optimize network performance.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→										PSO	PSO
↓ Course Outcomes										1	2
25MCA103.1										2	-
25MCA103.2										2	-
25MCA103.3										2	-

	25MCA103.4	2	3	2	-	-	-	-	3	1	-
	25MCA103.5	2	3	2	-	-	-	-	3	1	-
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	William Stallings: Data and Computer Communication, 10 th Edition, Pearson Education, 2022										
2.	Forouzan B. A., Data Communications and Networking (6e), Tata McGraw Hill 2022.										
3.	Computer Networks, Andrew S. Tanenbaum & David J. Wetherall.										
REFERENCE BOOKS:											
1.	James F. Kurose & Keith W. Ross – Computer Networking: A Top-Down Approach (8 th Edition, Pearson)										
2.	Douglas E. Comer – Internetworking with TCP/IP: Principles, Protocols, and Architecture										
3.	Larry Peterson & Bruce Davie – Computer Networks: A Systems Approach (6 th Edition, Morgan Kaufmann)										
E Books / MOOCs/ NPTEL:											
1.	NPTEL- https://onlinecourses.nptel.ac.in/noc25_cs15/preview										
2.	https://freecomputerbooks.com/An-Introduction-to-Computer-Networks-by-Peter-Dordal.html										
3.	NPTEL - https://onlinecourses.nptel.ac.in/noc25_cs15/preview										

MATHEMATICAL FOUNDATION FOR COMPUTER APPLICATIONS				
Course Code		25MCA104	Course Type	PCC
Teaching Hours/Week (L: T: P: S)		3:1:0:0	Credits	04
Total Teaching Hours		50	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications				
Course Objectives:				
1.	Understand operations on set theory, Solve example problems on set operations			
2.	Understand and solve problems on addition subtraction and multiplication of matrices.			
3.	Solve problems on eigen values and eigen vectors			
4.	Find mean, variance and covariance of two discrete and continuous random variable			
5.	Solve problems related to Graphs			
UNIT-I				
Set theory and operations on sets				05 Hours
Introduction, Venn Diagrams, subsets, The size of a set, power sets, cartesian products, Using set notations with quantifiers, Truth sets and quantifies				
Operations on sets : Introduction, set identities, generalizes unions and intersections, computer representations of sets				
Cardinality, Matrices, Eigen values and eigen vectors				05 Hours
Cardinality : Introduction, countable sets, an uncountable set				
Inclusion - exclusion principle : Introduction, the principle of inclusion exclusion				
Matrices : Introduction, matrix arithmetic, transposes and powers of matrices, Zero-one matrices				
Eigen values and eigen vectors :				
Finding Eigen values and Eigen vectors : Introduction, Procedure to find Eigen values, Procedure to find Eigen Vectors				
UNIT-II				
Propositional Logic, Applications of Propositional Logic				05 Hours
Propositional Logic : Introduction, Propositions, Conditional statements, Converse, Contrapositive and Inverse statements, Biconditional statements, Truth tables of Compound propositions, Precedence of logical operators, logic and bit operations				
Applications of Propositional Logic :				
Introduction, Translating English sentences, System Specifications, Logic Circuits, Logic Puzzles				
Propositional Equivalences, Predicates and Quantifiers				05 Hours
Propositional Equivalences :				
Introduction, Logical Equivalences, Using De Morgan's Laws, Constructing new logical equivalences, Propositional satisfiability, solving satisfiability problems				
Predicates and Quantifiers :				
Introduction, Predicates, Preconditions and post conditions, Quantifiers : Universal quantifier and existential quantifiers and uniqueness quantifiers				
UNIT-III				
Relations and their Properties, n-ary Relations and their Applications				05 Hours
Relations and their Properties :				
Functions as relations, relations on a set, properties of relations, combining relations				
n-ary Relations and their Applications :				
Introduction, n-ary relations, databases and relations, operations on n-ary relations, SQL				

Representing Relations, Closures of Relations, Equivalence relations										05 Hours	
Representing Relations : Introduction, representing relations using Matrices, representing using diagraphs											
Closures of Relations : Introduction, closures, paths in directed graphs, transitive closures, Warshall's algorithm.											
Equivalence relations : Introduction, Equivalence relations, equivalence classes, equivalence classes and partitions											
UNIT-IV											
The concept of probability, The axioms of probability										05 Hours	
The concept of probability : Introduction, Sample space and events											
The axioms of probability : Introduction, Properties of Probability, Bayes' Theorem, Basics of Probability, Concept of random variable											
Probability Distributions										05 Hours	
Discrete probability distributions : Introduction, Probability distribution for discrete random variable, expected values, Mean, variance and co-variance											
Continuous probability distributions : Introduction, Probability density function, expected values, Mean, variance and co-variance Binomial and normal distribution											
UNIT-V											
Graph Models Graph Terminology and Special Types of Graphs										05 Hours	
Introduction, Basic Terminology, some special simple graphs, Bipartite Graphs, Bipartite Graphs and matchings, some applications of special types of graphs, new graphs from old											
Representing Graphs and Graph Isomorphism, Euler and Hamilton Paths										05 Hours	
Representing Graphs and Graph Isomorphism : Introduction, Representing Graphs, Adjacency Matrices, Incidence Matrices, Isomorphism of graphs, Determining whether two simple graphs are isomorphic											
Euler and Hamilton Paths : Introduction, Euler paths and circuits, Hamilton paths and circuits, Applications of Hamilton Circuits											
Course Outcomes: At the end of the course student will be able to											
1.	Solve problems effectively on set operations.										
2.	Illustrate and solve problems on addition subtraction and multiplication of matrices.										
3.	Outline translation of sentences based on mathematical logic and solve problems related to applications of mathematical logic.										
4.	Find mean, variance and covariance of discrete and continuous random variable.										
5.	Demonstrate and solve problems related to Graphs.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2

	25MCA104.1	3	3	3	3	1	1	3	3	3	2	
	25MCA104.2	3	3	3	3	1	1	3	3	3	3	
	25MCA104.3	3	3	3	3	1	1	3	3	3	3	
	25MCA104.4	3	3	3	3	1	1	3	3	3	3	
	25MCA104.5	3	3	3	3	1	1	3	3	3	2	
1: Low 2: Medium 3: High												
TEXTBOOKS:												
1.	Kenneth H Rosen, “Discrete Mathematics and its Applications”, McGraw Hill Publications, 8 th Edition, 2021.											
2.	Wolpole Myers Ye “Probability and Statistics for engineers and Scientist” Pearson Education, 9 th Edition, 2024											
REFERENCE BOOKS:												
1.	Richard A Johnson and C.B Gupta “Probability and statistics for engineers” Pearson Education, 9 th Edition.											
2.	J.K Sharma “Discrete Mathematics”, Mac Millian Publishers India, 4 th Edition, 2022.											
E Books / MOOCs/ NPTEL												
1.	H. Pishro-Nik, "Introduction to probability, statistics, and random processes", available at https://www.probabilitycourse.com , Kappa Research LLC, 2014.											

RESEARCH METHODOLOGY AND PUBLICATION ETHICS				
Course Code		25MCA105	Course Type	PCC
Teaching Hours/Week (L:T:P:S)		3:0:0:0	Credits	03
Total Teaching Hours		40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications				
Course Objectives:				
1.	To understand the Meaning, Objectives and Characteristics of research			
2.	To understand how to select and define the research problem effectively.			
3.	To explore Research design and methodologies and data collection techniques			
4.	To analyze ethical considerations, research integrity, and plagiarism detection.			
5.	To understand Intellectual Property Rights and scientific publication practices.			
UNIT - I				
Introduction to Research:				08 Hours
Meaning, Objectives and Characteristics of research, Research methods Vs Methodology, Types of research, Descriptive Vs. Analytical, Applied Vs. Fundamental, Quantitative Vs. Qualitative, Conceptual Vs. Empirical, Research process, Criteria of good research, Importance of AI and ML in Research Analysis.				
UNIT – II				
Research Problem Identification & Literature Review:				08 Hours
Developing a research plan. Defining and Selecting a Research Problem - Necessity of defining the problem -Techniques involved in defining the problem - Importance of literature review in defining a problem - Survey of literature - Primary and secondary sources ,Use of Digital Libraries, Google Scholar and AI-based Research Assistance (ChatGPT, Zotero,Mendeley)				
UNIT – III				
Research design, Data Collection & Analysis:				08 Hours
Research design, Basic Principles, Need of research design, Features of good design, Important concepts relating to research design.				
Sampling design, Steps in sampling design, Sampling Techniques, Characteristics of a good sample design, Types of sample designs, Measurement and scaling techniques, Methods of data collection, Collection of primary data, Data collection instruments, Statistical Tools for Data Analysis.				
Unit – IV				
Ethics in Research & Scientific Writing:				08 Hours
Philosophy and Ethics, Scientific Conduct: Ethics: definition, moral philosophy, nature of moral judgments and reactions. Scientific Conduct Ethics with respect to science and research, Plagiarism and Detection Tools (Turnitin ,Grammarly), Intellectual Honesty and Research Integrity, Scientific Paper Writing, Journal Selection, and Peer Review Process.				
Unit - V				
Intellectual Property Rights & Research Publications:				08 Hours
Intellectual Property Rights(IPR) : Patent ,Copyrights, Trademarks ,and Trade Secrets. IPR Laws in India and International Context (TRIPS Agreement), Open Access Vs Paid Journals, Predatory Journals Awareness. Research Funding, Grant Proposal Writing and Patent Filing Process.				
Course Outcomes: At the end of the course student will be able to				

1.	Outline the Meaning, Objectives and Characteristics of research.
2.	Demonstrate how to select and define a research problem with a literature review.
3.	Design appropriate research methodologies and analyze data effectively.
4.	Assess ethical considerations, plagiarism detection and research integrity.
5.	Demonstrate knowledge of intellectual property rights and scientific publishing

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA105.1	3	2	1	1	-	-	-	2	2	1
25MCA105.2	2	3	2	3	-	-	-	2	3	2
25MCA105.3	2	2	3	3	-	-	-	2	3	3
25MCA105.4	-	-	-	-	-	-	3	2	2	3
25MCA105.5	-	-	-	-	-	3	2	3	3	2

1: Low 2: Medium 3: High
TEXT BOOKS:

1.	Kothari, C. R., 1990. Research Methodology : Methods and Techniques. New Age International. 418 p.
2.	Dipankar Deb • Rajeeb Dey, Valentina E. Balas “Engineering Research Methodology”, ISSN 1868- 4394 ISSN 1868-4408 (electronic), Intelligent Systems Reference Library, ISBN 978-981-13- 2946-3 ISBN 978-981-13-2947-0 (eBook), https://doi.org/10.1007/978-981-13-2947-0

REFERENCE BOOKS:

1.	Sinha, S. C. and Dhiman, A. K., 2002. Research Methodology, EssEss Publications. 2 volumes.
2.	Garg, B.L., Karadia, R., Agarwal, F. and Agarwal, U.K., 2002. An Introduction to Research Methodology, RBSA Publishers.
3.	Anderson, T. W., An Introduction to Multivariate Statistical Analysis, Wiley Eastern Pvt., Ltd., New Delhi
4.	Day, R.A., 1992. How to Write and Publish a Scientific Paper, Cambridge University Press.
5.	Fink, A., 2009. Conducting Research Literature Reviews: From the Internet to Paper. Sage Publications
6.	Coley, S. M. and Scheinberg, C. A., 1990, "Proposal Writing", Sage Publications.
7.	Intellectual Property Rights in the Global Economy : Keith Eugene Maskus, Institute for International Economics, Washington, DC, 2000
8.	Subbarau N R – Handbook on Intellectual Property Law and Practice-S Viswanathan Printers and Publishing Private Limited. 1998

ADVANCED OPERATING SYSTEMS WITH UNIX			
Course Code	25MCA106	Course Type	PCC
Teaching Hours/Week (L: T: P:S)	3:1:0:0	Credits	04
Total Teaching Hours	50	CIE + SEE Marks	50+50
Teaching Department: MCA			
Course Objectives:			
1.	Understand Advanced Operating System Concepts		
2.	Gain In-Depth Knowledge of UNIX Internals		
3.	Analyze Distributed and Multiprocessor OS Mechanisms		
4.	Learn Security and Performance Optimization Techniques		
5.	Explore Emerging Trends in Operating Systems		
UNIT-I			
Fundamentals of Advanced Operating Systems:			05 Hours
Overview of Advanced Operating Systems. Types of Advanced OS: Distributed, Multiprocessor, Real-Time, Cloud-Based.			
Operating System Structure:			06 Hours
Process Synchronization & Communication (IPC, Semaphores, Monitors, Message Passing), Deadlock Detection, Avoidance, and Recovery, Case Study: Real-World Advanced Operating Systems (Unix, Linux,Android).			
UNIT-II			
UNIX Operating System Internals:			10 Hours
Evolution & Architecture of UNIX, Process Management in UNIX (Fork, Exec, Signals), File System in UNIX (Structure, Inodes, System Calls) .Memory Management & Virtual Memory in UNIX, UNIX System Performance & Optimization			
UNIT-III			
Distributed Operating Systems:			11 Hours
Introduction to Distributed Systems and Their Architecture, Distributed Process Scheduling & Load Balancing, Distributed File Systems (NFS, HDFS) & Remote Procedure Calls (RPC), Synchronization in Distributed Systems			
UNIT-IV			
Security and Performance in UNIX & Advanced OS			10 Hours
Security Mechanisms in UNIX (Access Control, Firewalls, Authentication), Intrusion Detection & Prevention, Malware Protection, Performance Optimization Techniques in UNIX, Resource Management & Scheduling in Multi-core Systems			
UNIT-V			
Emerging Trends and Case Studies in Operating Systems			08 Hours
Cloud-based & Virtualized Operating Systems (Docker, Kubernetes), Real-Time Operating Systems (RTOS) & Embedded OS, AI and Machine Learning Integration in OS.			
Course Outcomes: At the end of the course student will be able to			

1.	Demonstrate a Strong Understanding of Advanced OS Architectures.
2.	Analyze and Work with UNIX Operating System Internals.
3.	Develop an Understanding of Distributed and Multiprocessor Systems.
4.	Apply Security and Performance Optimization Techniques in UNIX.
5.	Evaluate Emerging Trends and Future Directions in Operating Systems.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA106.1	3	3	2	2	-	-	-	2	3	3
25MCA106.2	3	3	3	3	-	-	-	2	3	2
25MCA106.3	3	3	2	3	-	-	-	3	3	2
25MCA106.4	3	3	3	2	2	-	-	2	3	2
25MCA106.5	3	3	3	3	2	2	2	3	3	2

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Mukesh Singhal, Niranjan G. Shivaratri, Advanced Concepts in Operating Systems, McGraw-Hill -2017
2. Robert Love, Linux System Programming, 2nd edition, 2013
3. Andrew S. Tanenbaum, Modern Operating Systems, 4th Edition, Pearson, 2014
4. William Stallings, Operating Systems: Internals and Design Principles, 9th Edition, Pearson, 2019

REFERENCE BOOKS

1. Abraham Silberschatz, Operating System Concepts, 10th Edition, Wiley
1. UNIX: The Complete Reference: Kenneth Rosen et al, Osborne/McGraw Hill, 2010
2. Maurice J. Bach, *The Design of the UNIX Operating System*, Pearson
3. Simson Garfinkel, Gene Spafford, and Alan Schwartz, Practical UNIX and Internet Security, 3rd edition

E Books / MOOCs/ NPTEL

1. <https://www.oreilly.com/library/view/learning-the-unix/1565923901/>

DATA STRUCTURES WITH ALGORITHMS LAB			
Course Code	25MCA107	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	0:0:4:0	Credits	02
Total Teaching Hours	26	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Implement the primitive data structures such as arrays and structures.		
2.	Implement Stack, Queue data structures statically using arrays and dynamically using linked lists.		
3.	Implement recursive and non recursive programs.		
4.	Implement the non linear data structures such as Linked list, Binary Trees, Graphs.		
5.	Implement different sorting methods and searching methods		
List of Experiments			
1.	C programs using pointers.		
2.	C programs using Arrays and Structures.		
3.	C programs to demonstrate the operations of stack. (using arrays and structures).		
4.	C programs to convert the valid infix arithmetic expression to postfix and prefix form.		
5.	Evaluate postfix expression.		
6.	Recursive C programs to i) Find the Factorial of a number. ii) Find Fibonacci of a number. iii) Solve Towers of Hanoi Problem. iv) Find the GCD of two numbers. v) Find the multiplication of two numbers.		
7.	C programs to simulate the working of simple queue, circular queue and priority queue using arrays and structures.		
8.	C programs to demonstrate the operations of singly & doubly linked list. (operations like insert a node at the front, at the back, at the specified position; delete a node from the front end, from the back end, from the specified position; search a node, if the info field is specified; display all the nodes in the list).		
9.	Implement stack and queues dynamically using Linked lists.		
10.	C programs on binary trees. (Construct a binary search tree and Traverse the tree using all the methods i.e., inorder, preorder, postorder).		
11.	C programs to demonstrate the different searching techniques over a list of integers. (Linear search, Binary search).		
12.	C programs to sort a list of elements using different sorting techniques. (Bubble sort selection sort, quick sort, simple insertion sort, shell sort, merge sort, Heap sort, Binary tree sort).		
Course Outcomes: At the end of the course student will be able to			
1.	Implement static and dynamic Stack and Queue data structures. Demonstrate the concept of Recursive programs and the implementation details of linked lists.		
2.	Explain the implementation details of Binary tree and Graph data structures. Implement and Illustrate various Sorting and Searching Techniques		

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes										
Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA107.1	3	3	3	3	3	-	-	-	3	2
25MCA107.2	3	3	3	3	3	-	-	-	3	2
1: Low 2: Medium 3: High										
REFERENCE BOOKS:										
1.	Yedidyah Langsam and Moshe J Augenstein and Aaron M Tenenbaum: Data Structures using C and C++, 2 nd Edition, Pearson Education Asia, 2002.									
2.	Reema Thareja – "Data Structures Using C", 2 nd Edition, Oxford University press									
3.	Jean-Paul Tremblay: An Introduction to Data Structures with Applications, Tata McGraw Hill									
4.	Ellis Horowitz and Sartaj Sahni, Fundamentals of Data Structures in C, 2 nd Edition, Universities Press.									
5.	Introduction to the Design and Analysis of Algorithms. Anany Levitin, Pearson Education, 2 nd Edition.									
6.	Horowitz E., Sahani S., Rajashekharan S: Computer Algorithm, Galgotia Publications.									

ADVANCED DATABASE SYSTEMS LAB													
Course Code				25MCA108			Course Type			PCC			
Teaching Hours/Week (L: T: P:S)				0:0:4:0			Credits			02			
Total Teaching Hours				26			CIE + SEE Marks			50+50			
Teaching Department: Master of Computer Applications													
Course Objectives:													
1.	Create and query the databases using SQL DML, DDL and DQL commands.												
2.	Use the features of MySQL to write procedural programs including stored procedures, cursors, and triggers.												
3.	Create and query NoSQL database using Mongodb.												
Course Outcomes: At the end of the course students will be able to													
1.	Apply SQL and procedural programming concepts in SQL and MySQL to query, update, and manage relational databases effectively.												
2.	Develop CRUD operations in a NoSQL database like MongoDB to manage unstructured and semi-structured data.												
List of Experiments													
1.	Exercise on creating tables.												
2.	Exercise on altering tables, dropping tables.												
3.	Exercise on giving table level constraints, field level constraints.												
4.	Exercise on insertion, retrieval, deletion and modification of data values.												
5.	Exercise on ORDER BY, GROUP BY, HAVING clauses.												
6.	Exercise on Aggregate functions in SQL.												
7.	Exercise on Joins, Unions, Sub queries, Nested Sub queries, Co-related sub-queries.												
8.	Exercise on creating views, dropping views.												
9.	Exercise on MySQL : control structures, connecting tables.												
10.	Exercise on MySQL : Procedures, Cursors, Triggers												
11.	Exercise on MongoDB.												
Course Outcomes: At the end of the course students will be able to													
1.	Use database language statements to query, update, and manage a database using SQL and PL/SQL.												
2.	Apply the concepts used in performing CRUD operations using NoSQL database like MongoDB.												
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes													
Program Outcomes→				1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes												1	2
25MCA108.1				3	-	3	2	3	2	-	3	3	-
25MCA108.2				3	-	3	3	2	3	-	3	3	-
1: Low 2: Medium 3: High													
TEXTBOOKS:													
1.	Elmasri and Navathe: Fundamentals of Database Systems, 7 th Edition 2016												

2.	Ivan Bayross : Commercial Application Development using Oracle Developer 2000
3.	Seema Acharya, Subhashini Chellappan, “Big Data Analytics”, 1 st Edition, Wiley, 2015
4.	Raghu Ramakrishnan and Johannes Gehrke : Database Management Systems, 6 th Edition, McGraw-Hill
REFERENCE BOOKS	
1.	Silberschatz, Korth and Sudarshan: Database Systems Concepts, 6 th Edition, McGraw-Hill
2.	Alexis Leon, Mathews Leon: Database Management Systems, Vikas Publishing House
3.	Connolly: Database Systems: A Practical Approach To Design Implementation And Management, 3 rd Edition, Person Education
E Books / MOOCs/ NPTEL	
1.	NOC : Fundamentals of Database Systems(Course sponsored by Aricent), IIT Kanpur, Dr. Arnab Bhattacharya https://nptel.ac.in/courses/106104135
2.	Oracle PL/SQL Language Pocket Reference, 5th Edition: A Guide to Oracle's PL/SQL Language Fundamentals. https://www.pdfdrive.com/oracle-plsql-language-pocket-reference-5th-edition-a_guide-to-oracles-plsql-language-fundamentals-d167357800.html
3.	MySQL® Notes for Professionals book: https://www.goalkicker.com/MySQLBook/
4.	Learning MongoDB: https://www.dbooks.org/learning-mongodb-5605980453/#google_vignette

DATA COMMUNICATION AND NETWORKS LAB													
Course Code				25MCA109				Course Type				PCC	
Teaching Hours/Week (L: T: P: S)				0:0:4:0				Credits				02	
Total Teaching Hours				26				CIE + SEE Marks				50+50	
Teaching Department: Master of Computer Applications													
Course Objectives:													
1.		To get practical knowledge of working principles of various communication protocols											
2.		To analyze the various routing algorithms and To know the concept of data/packet transfer between nodes.											
3.		Analyse structure and formats of TCP/IP layer protocols using network tools like network simulators.											
4.		Implementing various network algorithms such as error control, error detection, routing and security related algorithms.											
List of Experiments													
1.		Implementation of error detection mechanisms											
2.		Implementation of the congestion control algorithms.											
3.		Implementation of routing algorithms.											
4.		Implementation of client / server programs using TCP and UDP.											
5.		Implementation of RSA algorithm for encryption/Decryption of message between client & server.											
6.		Implementation of frame sorting logic, to order frames received from multiple senders.											
7.		Implementation of TELNET.											
8.		Simulation of Networking components using any Network Simulation software											
9.		Implementation of wireless technology											
10.		Use of Open NMS and TCL tool.											
Course Outcomes: At the end of the course student will be able to													
1.		Interpret the practical approach to network communication protocols.											
2.		Classify the applications of various Routing Protocols/Algorithms.											
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes													
Program Outcomes→				1	2	3	4	5	6	7	8	PSO	PSO
↓Course Outcomes												1	2
25MCA109.1				3	3	3	3	2	-	3	3	3	1
25MCA109.2				3	3	2	2	3	-	3	3	3	1
1: Low 2: Medium 3: High													
REFERENCE BOOKS:													
1.		Cisco CCNA Study Guide v2.71, Aaron Balchunas											
2.		CCNA, Study Guide, 6 th Edition, TodLammle											
E Resources													
1.		https://www.isi.edu/nsnam/ns/											
2.		https://www.geeksforgeeks.org/network-simulator-3/											